

INFO6007
Project Management in IT

Week 03: Scope Management
Plan and Time Management

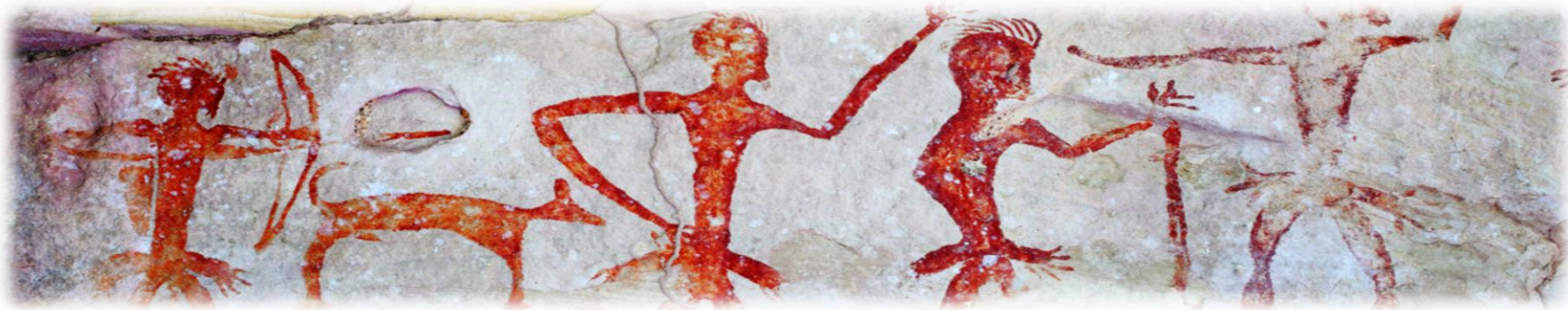
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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Agenda

- Introduction
- Project Scope Management
- Project Time Management

Project Scope Management

Introduction

- A Scope Management Plan is a crucial component of a project management plan that defines how the project scope will be defined, validated, and controlled.
- It ensures that all project work, and only that work, is included and completed successfully.
- A Scope Management Plan outlines the processes and procedures for managing the project scope throughout the project lifecycle.

Key components of Scope Management Plan

Component	Description
Scope Statement	High-level narrative of what the project will deliver (and won't).
Scope Management Approach	Methodology used to define and manage scope (e.g., Agile, Waterfall).
Requirement Collection	How stakeholder needs will be gathered and documented.
Work Breakdown Structure	A hierarchical decomposition of the project deliverables.
Scope Validation Process	How deliverables will be reviewed and approved by stakeholders.
Scope Control Process	How changes to the scope will be identified, assessed, and managed.
Roles and Responsibilities	Who is responsible for defining, validating, and controlling scope.
Assumptions and Constraints	Any conditions or limitations considered in defining the project scope.

Project Scope Management Overview

5.1 Plan Scope Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Scope management plan
 - .2 Requirements management plan

5.4 Create WBS

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Decomposition
- .3 Outputs
 - .1 Scope baseline
 - .2 Project documents updates

5.2 Collect Requirements

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Business documents
 - .5 Agreements
 - .6 Enterprise environmental factors
 - .7 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Data analysis
 - .4 Decision making
 - .5 Data representation
 - .6 Interpersonal and team skills
 - .7 Context diagram
 - .8 Prototypes
- .3 Outputs
 - .1 Requirements documentation
 - .2 Requirements traceability matrix

5.5 Validate Scope

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Verified deliverables
 - .4 Work performance data
- .2 Tools & Techniques
 - .1 Inspection
 - .2 Decision making
- .3 Outputs
 - .1 Accepted deliverables
 - .2 Work performance information
 - .3 Change requests
 - .4 Project documents updates

5.3 Define Scope

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Decision making
 - .4 Interpersonal and team skills
 - .5 Product analysis
- .3 Outputs
 - .1 Project scope statement
 - .2 Project documents updates

5.6 Control Scope

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Data analysis
- .3 Outputs
 - .1 Work performance information
 - .2 Change requests
 - .3 Project management plan updates
 - .4 Project documents updates

6 main processes in Project Scope Management

Figure 5-1. Project Scope Management Overview

Case Study: Google Cloud Expansion Project

Mapping Scope Management Processes

Scenario: **Google Cloud's Global Data Center Expansion**

Company: **Google**

- **Project:**

Google Cloud plans to expand its data center infrastructure across new global locations to meet increasing demand for cloud computing and AI services. The project involves building new data centers, integrating advanced cooling systems, and ensuring compliance with global regulations. To manage this large-scale IT project effectively, Project Scope Management is applied using the six key processes.

1. Plan Scope Management

- Main objective is to create a scope management plan that outlines how the project scope will be defined, validated, and controlled. It ensures consistency and clarity throughout the project.
- Sets the foundation for all other scope-related decisions. Without this plan, scope-related efforts can become disorganized and inconsistent.
- Inputs:
 - Project charter, Project management plan, Enterprise environmental factors, and Organisational process assets.

5.1 Plan Scope Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Scope management plan
 - .2 Requirements management plan

2. Collecting Requirements

- Main objective is to determine, document, and manage stakeholder needs, expectations, and assumptions to meet the project objectives.

Techniques	Description
Interviews	One-on-one conversation to uncover needs.
Workshops	Interactive sessions for shared understanding.
Surveys/Questionnaires	Reach large stakeholder groups efficiently.
Prototyping	Early versions to refine requirements.
Observation	Studying users in action.

- Outputs - Requirements documentation, and Requirement Traceability Matrix.

5.2 Collect Requirements

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Business documents
 - .5 Agreements
 - .6 Enterprise environmental factors
 - .7 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data gathering
 - .3 Data analysis
 - .4 Decision making
 - .5 Data representation
 - .6 Interpersonal and team skills
 - .7 Context diagram
 - .8 Prototypes
- .3 Outputs
 - .1 Requirements documentation
 - .2 Requirements traceability matrix

Requirement Traceability Matrix

Project	Project Manager
Website Launch	David Albert
Contact	Date
555-555-5555	27-Jan-25

Requirements Traceability Matrix Example



ID	Requirement Name	Requirement Description	Type of Requirement	Requirement Source	Priority Level	Potential Risks	Requirement Status	Requirement Owner	Acceptance Criteria	Test Cases	Objectives and Goals	Requested By	Dept.
1	User Authentication	Enable secure login for users with unique credentials	Functional	Stakeholder Request	High	Unauthorized access, poor user adoption	In Progress	IT Team Lead	Users can log in and out securely	Test invalid login, password reset, and lockout scenarios	Improve platform security and user experience	Sam Heard	IT
2	Payment Integration	Implement payment gateway for seamless transactions	Functional	Market Analysis	High	Payment failures, loss of revenue	Not Started	Product Manager	Transactions are processed within 5 seconds	Test successful, failed, and duplicate payments	Enhance revenue stream and user convenience	Penny Wenn	Product Management
3	Data Backup Mechanism	Implement daily automatic backups of user data	Non-functional	Compliance Requirement	Medium	Data loss during recovery, increased storage cost	Completed	DevOps Engineer	Daily backups are accessible and restorable	Test backup scheduling, accessibility, and data recovery	Ensure data integrity and disaster recovery		IT

3. Defining Scope

- The main objective of this process is to develop a detailed project scope statement that includes deliverables, boundaries, acceptance criteria, and assumptions and constraints.
- The output of this process is a Project Scope statement.
- A project Scope statement includes product scope description, deliverables, exclusions, and criteria for success.
- This document becomes the official agreement of what is and isn't included in the project, reducing ambiguity.

5.3 Define Scope

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Project documents
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Decision making
 - .4 Interpersonal and team skills
 - .5 Product analysis
- .3 Outputs
 - .1 Project scope statement
 - .2 Project documents updates

4. Creating WBS

- The main purpose of this process is to break the project scope and deliverables into smaller, more manageable components.
- A WBS is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project.
- WBS is the backbone of project planning. It clarifies deliverables, aids in resource planning, and sets up cost/schedule tracking.
- The technique used to create a WBS is called decomposition. It is the process of breaking high-level deliverables into sub-deliverables, and then into work packages.

5.4 Create WBS

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Decomposition
- .3 Outputs
 - .1 Scope baseline
 - .2 Project documents updates

Approaches to developing the WBS

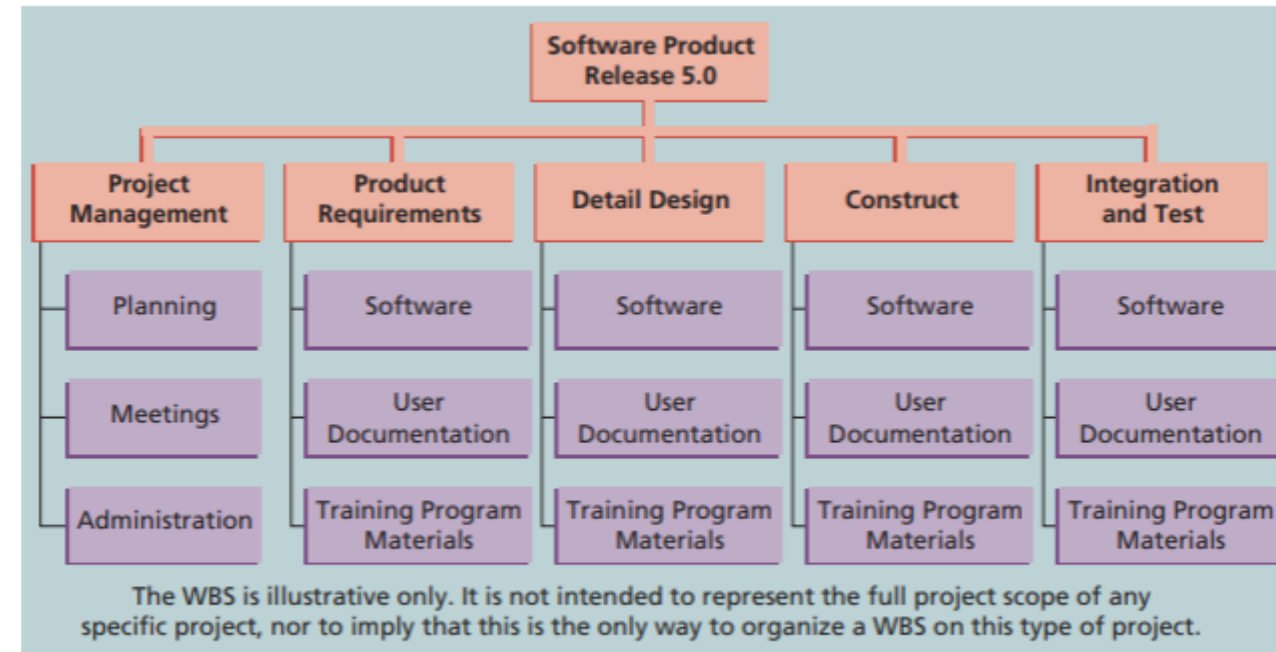
- **Using guidelines:** Some organisations, like the Department of Defense (DOD) provides guidelines for preparing WBS.
- **Analogy approach:** Review WBS of similar projects and tailor to your project.
- **Top-Down approach:** Start with the largest items of the project and break them down.
- **Bottom-Up approach:** Start with the specific tasks and roll them up.
- **Mind-mapping:** Uses branches radiating out from a core idea to structure thoughts and ideas.

Types of WBS

TABLE 5-6 Sample WBS dictionary entry

WBS Dictionary Entry March 20
Project Title: Information Technology (IT) Upgrade Project
WBS Item Number: 2.2
WBS Item Name: Database Update
Description: The IT department maintains an online database of hardware and software on the corporate intranet. We need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Physical Inventory, and must precede WBS Item Number 3.0, Hardware and Software Acquisition.

WBS Dictionary



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WBS Tree Diagram

Things to remember when creating the WBS

- A unit of work should appear at only one place in the WBS.
- The work content of a WBS item is the sum of the WBS items below it.
- A WBS item is the responsibility of only one individual, even though many people may be working on it.
- The WBS must be consistent with the way in which work is actually going to be performed.
- Project team members should be involved in developing the WBS to ensure consistency and buy-in.
- Each WBS item must be documented in the WBS dictionary.

5. Validating Scope

- The main objective is to gain formal acceptance of the completed project deliverables by the customer or sponsor.
- Major activities involves presenting deliverables for stakeholder inspection, gathering approvals, and managing feedback and documenting acceptance.
- It is important to validate the scope because without formal validation, disputes may arise later. This is also where scope creep can be caught before it escalates.
- The process produces the following output: Accepted deliverables, change requests (if the deliverable is not accepted), and work performance information.

5.5 Validate Scope

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Verified deliverables
 - .4 Work performance data
- .2 Tools & Techniques
 - .1 Inspection
 - .2 Decision making
- .3 Outputs
 - .1 Accepted deliverables
 - .2 Work performance information
 - .3 Change requests
 - .4 Project documents updates

6. Controlling Scope

- The main objective is to monitor the status of the project and product scope and manage changes to the scope baseline.
- The main techniques used to control scope are variance analysis (actual vs planned scope), performance reviews, and/or integrated change control process.
- This process will have the following outputs: Change requests, updated project documents, and work performance information.
- This process is important as it helps prevent scope creep, ensures work aligns with original plan, and supports accountability.

5.6 Control Scope

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Data analysis
- .3 Outputs
 - .1 Work performance information
 - .2 Change requests
 - .3 Project management plan updates
 - .4 Project documents updates

Scope Management Issues

Issue	Description	Impact
Scope Creep	Uncontrolled changes or continuous growth in a project's scope without adjustments to time, cost, and resources.	Delays, budget overruns, team burnout.
Lack of Stakeholder Engagement	Stakeholders are not properly involved in defining or validating scope.	Missed expectations, dissatisfaction, scope changes later.
Miscommunication	Between project team, client, or management regarding what is in or out of scope.	Confusion, conflict, inconsistent deliverables.
Overambitious Scope	Trying to achieve too much in one project cycle.	Increases complexity, risk, and chances of failure.

Mitigating the issues

- Clear Scope Statement with inclusions, exclusions, assumptions.
- Use a Requirements Traceability Matrix to link requirements to deliverables.
- Stakeholder Sign-Off on each stage of scope definition and change.
- Implement Formal Change Control Process to manage scope updates.
- Conduct Regular Scope Reviews to ensure alignment and control.

Project Schedule Management

Introduction

- **Project Schedule Management** is the process of planning, developing, managing, executing, and controlling a project's schedule so that it can be completed within the agreed time frame.
- In simpler terms – it is about **deciding what needs to be done, in what order, how long it will take, and making sure it stays on track** until the project is finished.

Key components of Schedule Management Plan

Component	Description
Schedule Development Approach	Methodology to be used (e.g., Critical Path Method, Agile, Rolling Wave Planning). Tools/software (MS Project, Primavera, Jira, etc.).
Activity Definition and Sequencing	How activities will be identified (WBS, deliverable decomposition). Rules for defining dependencies and constraints.
Estimating Durations	Techniques to be used (Analogous, Parametric, Three-point/Pert). Estimation assumptions and accuracy range.
Schedule Baseline	How and when the baseline will be set. Process for baseline approval.
Schedule Control and Monitoring	Performance metrics (milestone tracking). Frequency of progress updates. Variance thresholds that trigger corrective actions.
Reporting Format	How the schedule will be presented (Gantt charts, milestone charts, dashboards). Update frequency for stakeholders.

Project Schedule Management Overview

6.1 Plan Schedule Management

- .1 Inputs
 - .1 Project charter
 - .2 Project management plan
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Data analysis
 - .3 Meetings
- .3 Outputs
 - .1 Schedule management plan

6.4 Estimate Activity Durations

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Analogous estimating
 - .3 Parametric estimating
 - .4 Three-point estimating
 - .5 Bottom-up estimating
 - .6 Data analysis
 - .7 Decision making
 - .8 Meetings
- .3 Outputs
 - .1 Duration estimates
 - .2 Basis of estimates
 - .3 Project documents updates

6.2 Define Activities

- .1 Inputs
 - .1 Project management plan
 - .2 Enterprise environmental factors
 - .3 Organizational process assets
- .2 Tools & Techniques
 - .1 Expert judgment
 - .2 Decomposition
 - .3 Rolling wave planning
 - .4 Meetings
- .3 Outputs
 - .1 Activity list
 - .2 Activity attributes
 - .3 Milestone list
 - .4 Change requests
 - .5 Project management plan updates

6.5 Develop Schedule

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Agreements
 - .4 Enterprise environmental factors
 - .5 Organizational process assets
- .2 Tools & Techniques
 - .1 Schedule network analysis
 - .2 Critical path method
 - .3 Resource optimization
 - .4 Data analysis
 - .5 Leads and lags
 - .6 Schedule compression
 - .7 Project management information system
 - .8 Agile release planning
- .3 Outputs
 - .1 Schedule baseline
 - .2 Project schedule
 - .3 Schedule data
 - .4 Project calendars
 - .5 Change requests
 - .6 Project management plan updates
 - .7 Project documents updates

6.3 Sequence Activities

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Enterprise environmental factors
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Precedence diagramming method
 - .2 Dependency determination and integration
 - .3 Leads and lags
 - .4 Project management information system
- .3 Outputs
 - .1 Project schedule network diagrams
 - .2 Project documents updates

6.6 Control Schedule

- .1 Inputs
 - .1 Project management plan
 - .2 Project documents
 - .3 Work performance data
 - .4 Organizational process assets
- .2 Tools & Techniques
 - .1 Data analysis
 - .2 Critical path method
 - .3 Project management information system
 - .4 Resource optimization
 - .5 Leads and lags
 - .6 Schedule compression
- .3 Outputs
 - .1 Work performance information
 - .2 Schedule forecasts
 - .3 Change requests
 - .4 Project management plan updates
 - .5 Project documents updates

6 main processes in Project Schedule Management Plan



Case Study: Google Cloud Expansion Project

Mapping Schedule Management Processes

Scenario: **Google Cloud's Global Data Center Expansion**

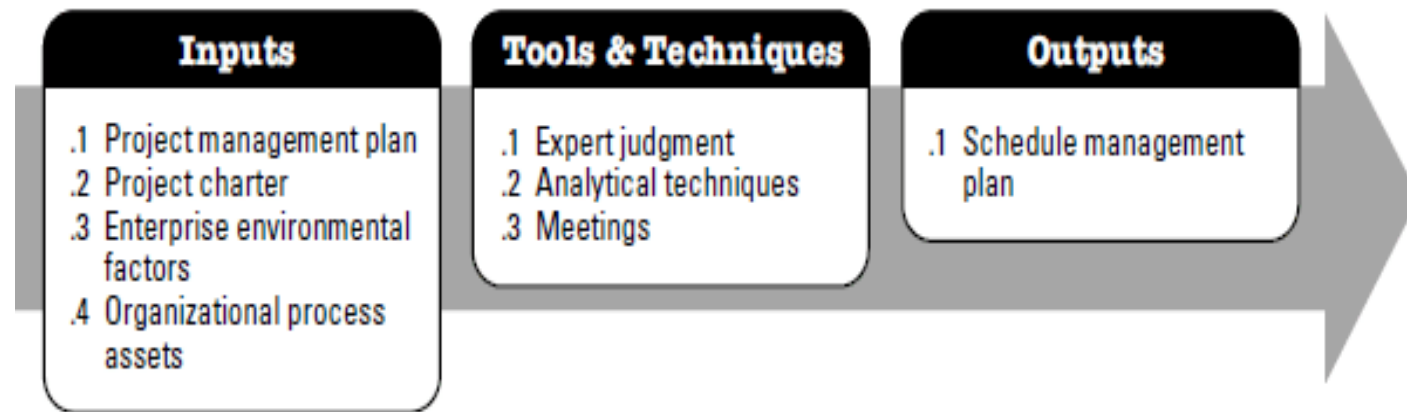
Company: **Google**

- **Project:**

Google Cloud is undertaking a global data center expansion to meet the surging demand for cloud computing and AI workloads. The initiative includes constructing new data centers in multiple continents, deploying advanced cooling technologies, and integrating high-speed network infrastructure. To ensure the project is delivered on time across diverse geographical regions, Project Schedule Management is applied using the six main processes.

1. Planning Schedule Management

- The main purpose of planning the schedule management plan is to define how the schedule will be created, monitored, and controlled.
- The key activities involved with planning includes:
 - Decide tools, techniques, and formats.
 - Assign responsibilities.
 - Identify performance metrics.

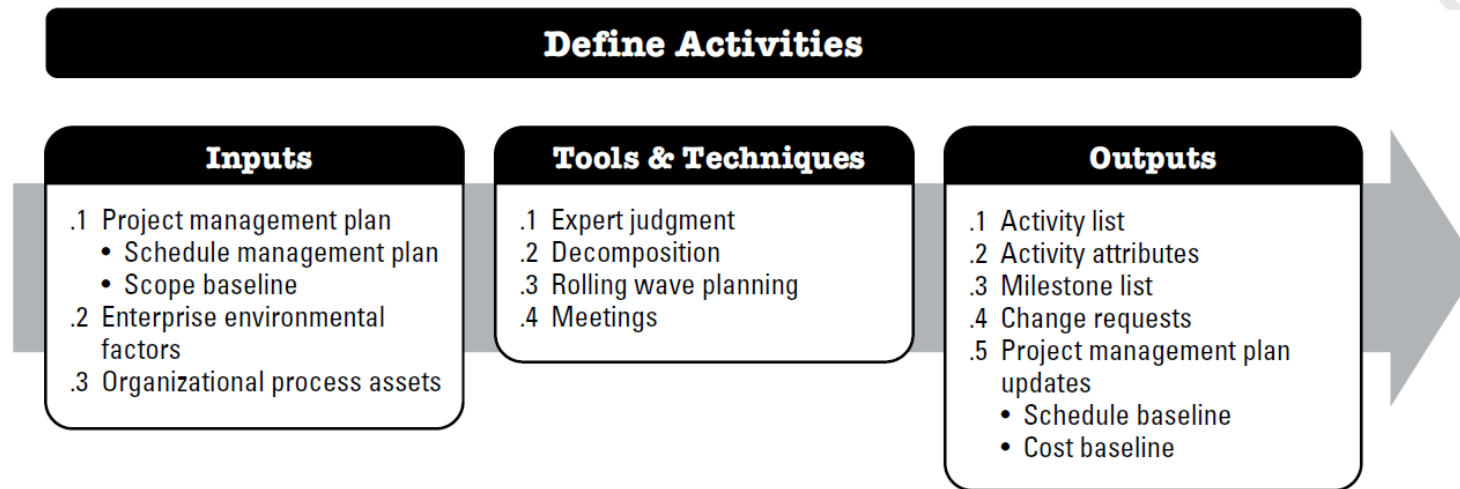


(Roseke, 2016)

Figure 6-3. Plan Schedule Management: Inputs, Tools & Techniques, and Outputs

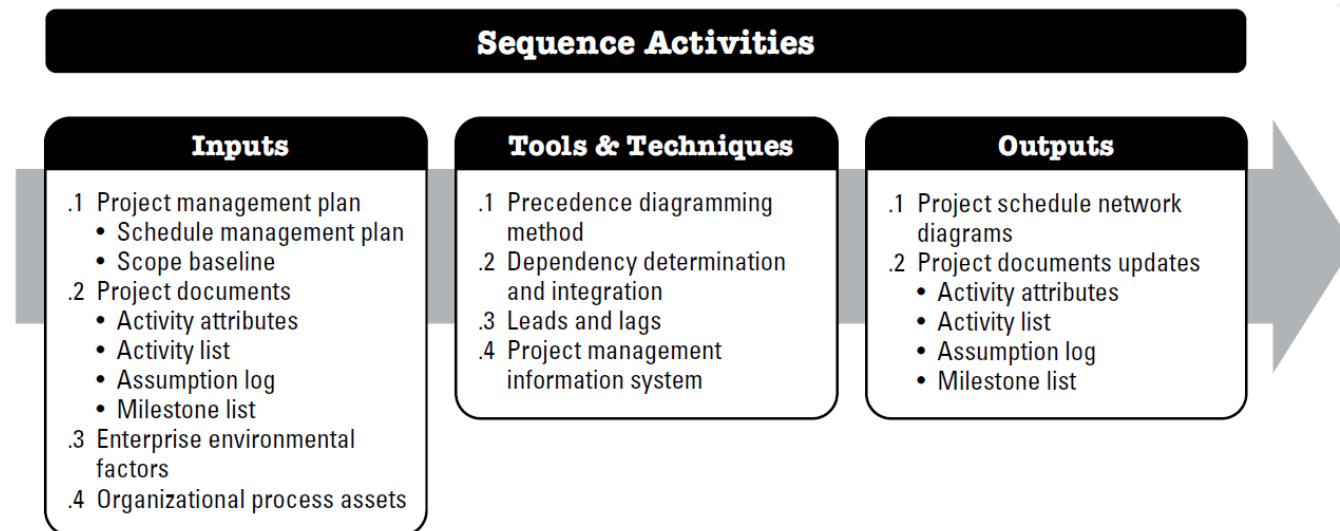
2. Defining Activities

- The main purpose of this process is to break down work packages from the Work Breakdown Structure (WBS) into specific, manageable tasks.
- Key activities in this process is:
 - Identify all activities required to produce deliverables.
 - Add activity attributes (duration, resources, predecessors).



3. Sequencing Activities

- The main purpose of this process is to determine the logical order of tasks and define dependencies.
- Key activities are:
 - Identify relationships (Finish-to-Start, Start-to-Start, etc.).
 - Create network diagrams.



(Roseke, 2016)

Figure 6-7. Sequence Activities: Inputs, Tools & Techniques, and Outputs

Four types of dependencies

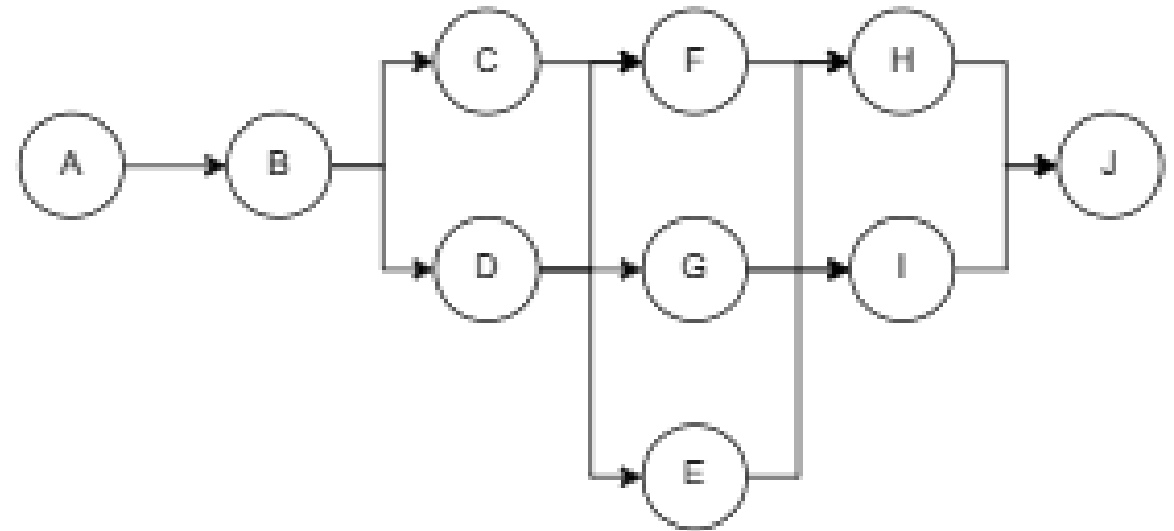
- Mandatory dependencies
 - Dependencies that are inherent to the work and cannot be changed; also called hard logic.
- Discretionary dependencies
 - Dependencies set based on best practices, preferences, or convenience; also called soft logic.
- External dependencies
 - Dependencies that rely on activities outside the project team's control.
- Internal dependencies
 - Dependencies between activities within the project team's control.

Network Diagram #1 – Activity on Node Diagram (AON)

- This a network diagram wherein the activities are represented by nodes (boxes) and the arrows shows dependencies between activities.
- Each box is an activity, and the arrows indicate the sequence or precedence relationships.
- This is the most widely used method today as it is easier to understand and draw.
- It is also supported by most project management softwares like MS Project and Primavera.

Example - Activity on Node Diagram

Activity	Estimated Duration	Predecessor
A	5	None
B	4	A
C	5	B
D	6	B
E	7	D
F	3	C, D
G	6	D
H	7	F, G
I	8	E, G
J	3	H, I

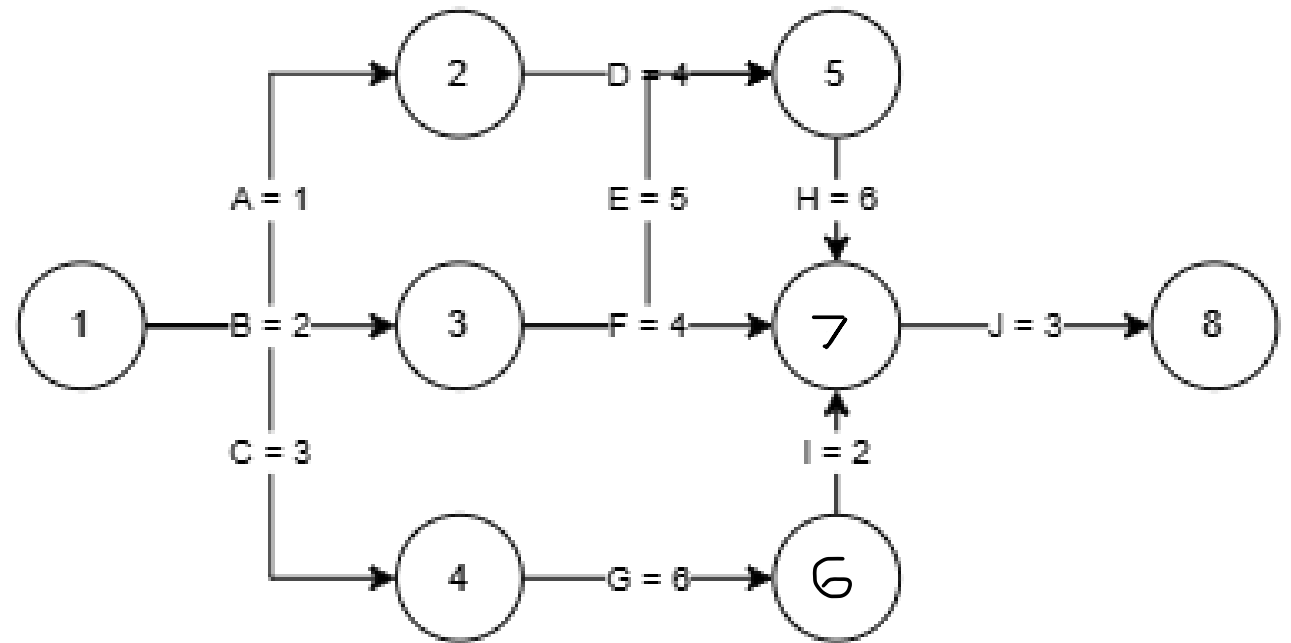


Network Diagram #2 – Activity on Arrow Diagram (AOA)

- A network diagram wherein activities are represented by arrows and the nodes represents events/milestones (start/end points).
- The arrow shows the activity duration and direction of workflow and the dependencies are shown by how arrows connect between nodes.
- It is less commonly used now due to complexity of the projects.
- This is also the only network diagram that usually requires dummy activities (arrows with 0 duration) to show dependencies properly.

Example - Activity on Arrow Diagram

Activity	Estimated Duration	Predecessor
A	1	None
B	2	None
C	3	None
D	4	A
E	5	B
F	4	B
G	6	C
H	6	D, E
I	2	G
J	3	F, H, I



Network Diagram #3 – Precedence Diagramming Method (PDM)

- It is a type of Activity on Node diagram that explicitly defines the different dependency types of activities. The dependency types are:
 - Finish to Start (FS) - Activity B starts after A finishes (most common).
 - Start to Start (SS) - Activity B starts when A starts.
 - Finish to Finish (FF) - Activity B finishes when A finishes.
 - Start to Finish (SF) - Activity B finishes when A starts (rare).
- This diagram shows more complex scheduling relationships and allows lead and lag times.
- This is used for more detailed and flexible schedule modelling and is supported by most modern PM tools.

Example – Precedence Diagramming Method

Task dependencies

The nature of the relationship between two linked tasks. You link tasks by defining a dependency between their finish and start dates. For example, the “Contact caterers” task must finish before the start of the “Determine menus” task. There are four kinds of task dependencies in Microsoft Project.

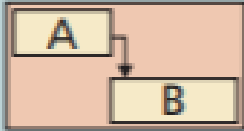
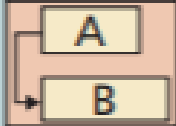
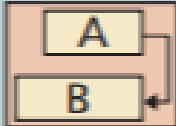
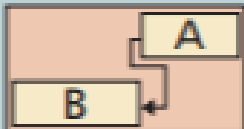
Task dependency	Example	Description
Finish-to-start (FS)		Task (B) cannot start until task (A) finishes.
Start-to-start (SS)		Task (B) cannot start until task (A) starts.
Finish-to-finish (FF)		Task (B) cannot finish until task (A) finishes.
Start-to-finish (SF)		Task (B) cannot finish until task (A) starts.

FIGURE 6-3 Task dependency types

4. Estimating Activity Durations

- Main purpose is to predict how long each activity will take, considering resource availability.
- Key activities are:
 - Use estimating techniques (Analogous, Parametric, Three-point/Pert, Expert Judgment).
 - Account for risks and uncertainties.

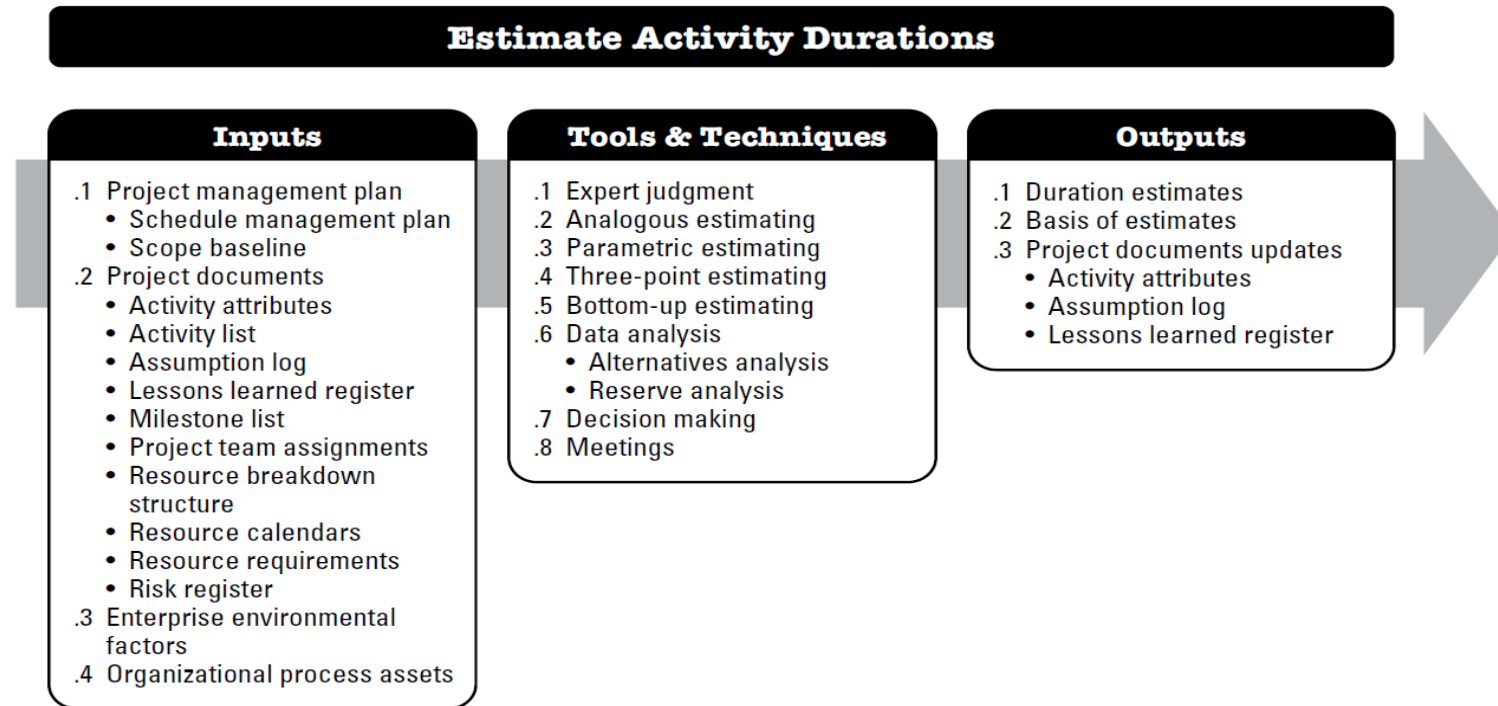


Figure 6-12. Estimate Activity Durations: Inputs, Tools & Techniques, and Outputs

(Roseke, 2016)

Three Point Estimate – PERT Analysis

- **PERT** is a method used to estimate how long a task will take when the time is uncertain. Instead of guessing one time, we estimate three possible times.
- The 3 estimates are:
 - **Optimistic (O)** - The shortest time in which the task can be completed if everything goes well.
 - **Most Likely (M)** - The best guess of the time required, assuming normal conditions.
 - **Pessimistic (P)** - The longest time the task might take if things go wrong.
- Formula is:
$$\text{PERT} = \frac{(O+4M+P)}{6}$$

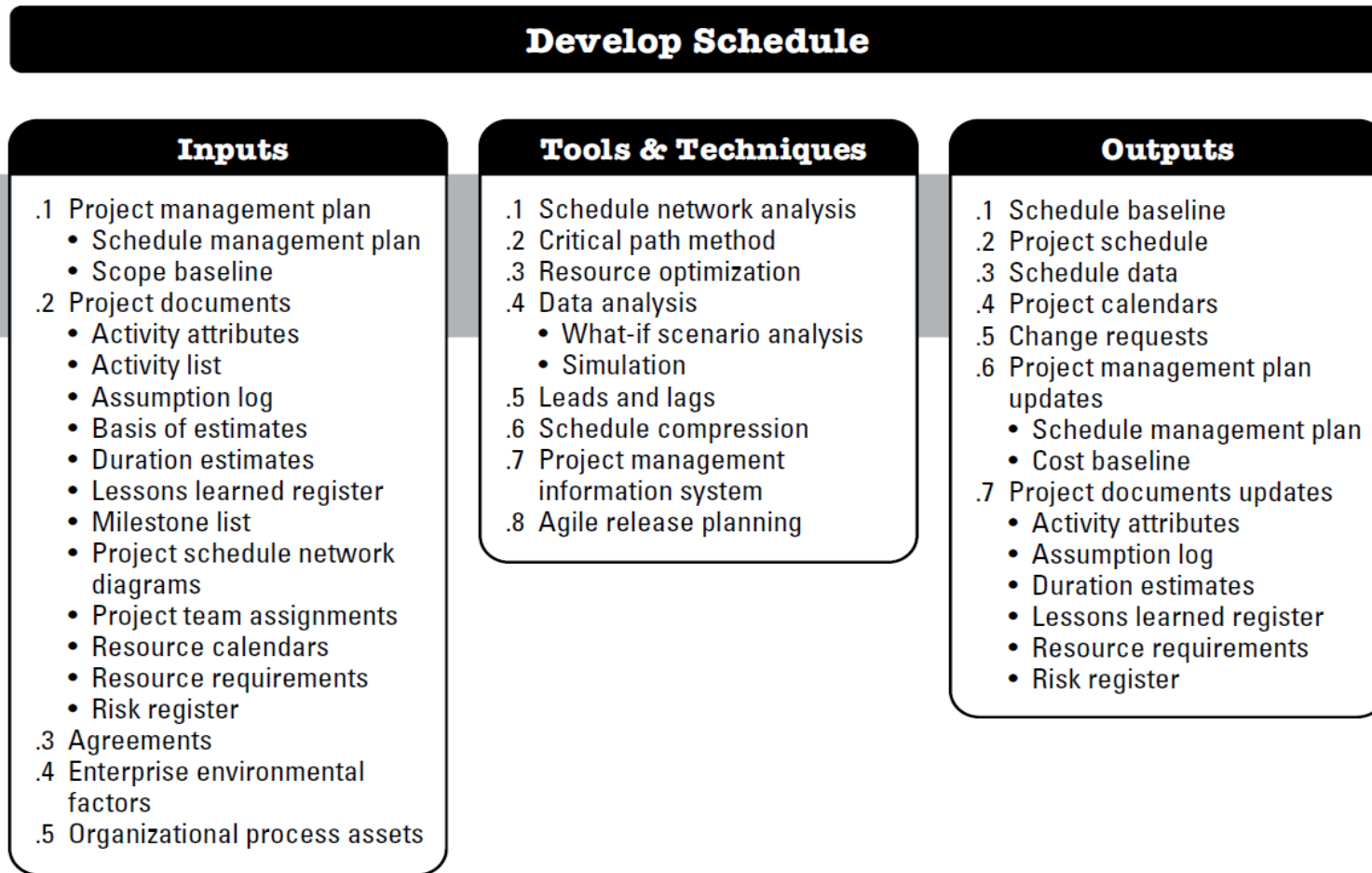
Analogous Estimating

- **Analogous Estimating** is a top-down project estimation technique where you predict the duration or cost of a task based on **historical data from similar projects or activities**.
- **Faster and less costly** than detailed estimating methods.
- **Requires expert judgment** to ensure similarities are valid.
- Works well in the **early stages** of a project when there's limited information.
- Can be used for **both time and cost estimates**.

Parametric Estimation

- **Parametric Estimation** is a quantitative project estimation technique where you use **a measurable relationship between variables** to calculate activity durations or costs.
- Uses **historical data + mathematical models**.
- Involves determining a **unit rate** and multiplying it by the number of required units.
- Works best when the relationship between variables is **consistent and scalable**.
- Can be applied to **time, cost, or resources**.

5. Developing Schedule



- Developing the schedule means creating the final project timeline.

- Key activities are:
 - Apply Critical Path Method (CPM).
 - Allocate resources and resolve conflicts.
 - Set the schedule baseline.

Figure 6-14. Develop Schedule: Inputs, Tools & Techniques, and Outputs

(Roseke, 2016)

Gantt Chart

- A **Gantt Chart** is one of the most common and visual tools in project schedule management.
- A **Gantt Chart** is a **bar chart** that displays project activities or tasks against time. Each task is represented by a horizontal bar whose length represents its duration.
- Uses:
 - Visualizing the project timeline.
 - Tracking progress against the baseline schedule.
 - Identifying overlaps, leads, and lags.
 - Communicating the schedule to stakeholders clearly.



ALL

TASK NAME

JUL, 24 '22

JUL, 31 '22

AUG, 7 '22

AUG, 14 '22

AUG, 21 '22

AUG, 28 '22

SEP, 4 '22

W T F S S M T W T F S S M T W T F S S M T W T F S S M T W T

1

Contracts

Contracts 100%



2

Proposals

Proposals 100% Mike Smith (Sample)



3

Documents Review

Documents Review 100% Mike Smith (Sample)



4

Bid Date

Bid Date 100% Mike Smith (Sample)

5

Award Date

8/2/2022

6

Design

Design 80%

7

Feasibility Study

Feasibility Study 100% Jennifer Jones (Sample)

8

Apply for Permits

Apply for Permits 100% Jennifer Jones (Sample)

9

Start Design Work

Start Design Work 75% Jennifer Jones (Sample)

10

Complete Design Work

8/22/2022

11

Procurement

Procurement 19%

12

Order Equipment

Order Equipment 0% Sam Watson (Sample)

13

Order Materials

Order Materials 0% Sam Watson (Sample)

14

Hire Crew

Hire Crew 25% Sam Watson (Sample)

+

15

Construction

16

Prep/Pre-construction

Prep/Pre-construction 0%

17

Construction Start Date

Construction Start Date 0%

18

Site work



Critical Path Method

- The **Critical Path Method** is a scheduling technique used to identify the **longest sequence of dependent activities** in a project, the path that determines the shortest possible project duration.
- Any delay on the Critical Path will DELAY the entire project.
- Advantages:
 - Helps prioritize tasks.
 - Identifies where flexibility exists.
 - Useful for schedule compression (Fast Tracking, Crashing).

6. Controlling Schedule

- Main purpose of this step is to monitor progress, identify variances, and take corrective actions.
- Key activities are:
 - Compare actual vs. planned performance.
 - Use Earned Value Management (EVM) metrics like SPI & SV.
 - Apply schedule compression techniques (Fast Tracking, Crashing) if needed.

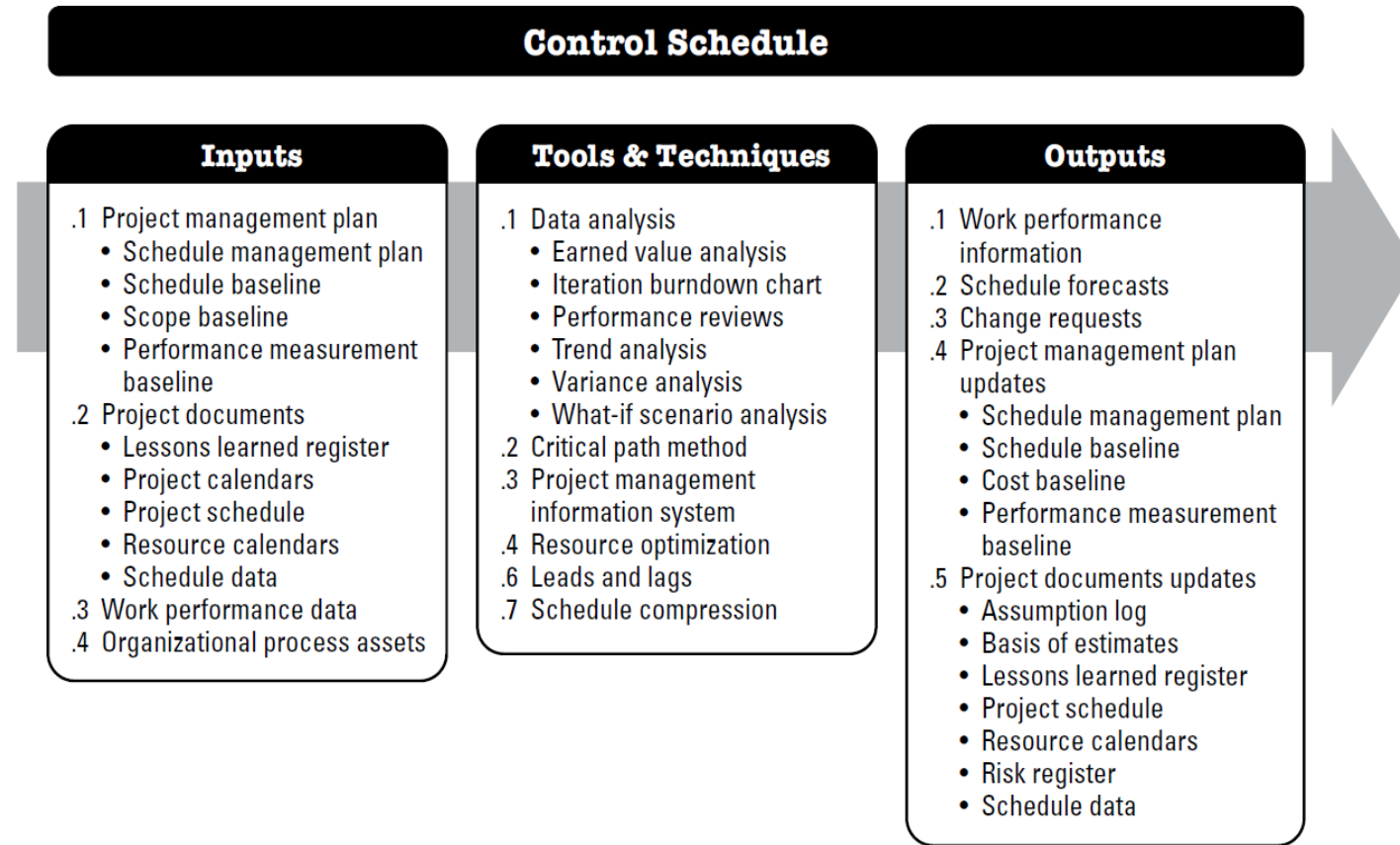


Figure 6-22. Control Schedule: Inputs, Tools & Techniques, and Outputs

(Roseke, 2016)

Schedule Compression

- Despite there being multiple ways of controlling the schedule, there are 2 very important concepts:
- Fast Tracking
 - Perform activities in parallel that were originally sequential.
 - Increases risk but shortens schedule.
- Crashing
 - Add more resources to critical path activities to finish sooner.
 - Increases cost, may increase risk.

Scope and Schedule Management in Different Development Approaches

Approach	Scope Management	Schedule Management
Waterfall	Scope is clearly defined at the beginning and changes are difficult.	Full project schedule is planned upfront with fixed phases.
Agile	Scope is flexible and evolves through backlog and user stories.	Work is done in short sprints (e.g., 2 weeks).
DevOps (Practice)	Scope continuously evolves with small improvements and updates.	Work follows continuous development, integration, and delivery cycles.

References

- Roseke, B., 2016. *Schedule Planning*. [Online]
Available at: <https://www.projectengineer.net/schedule-planning/>
[Accessed 12 08 2025].
- Schwalbe, K., 2018. Project Schedule Management. In: *Information Technology Project Management*. Boston: Cengage, pp. 241-284.

For You!

End of Lecture Questions

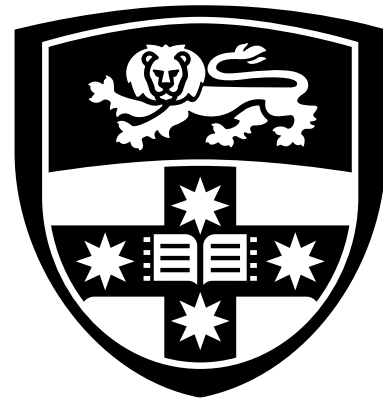
- Name and briefly explain the **six processes** of project scope management.
- Why is a **Work Breakdown Structure (WBS)** important in scope management?
- What are **scope creep** and its main causes?
- List the **six processes** in project schedule management.
- Explain the difference between **mandatory** and **discretionary dependencies**.
- What is the formula for **three-point estimating** in PERT analysis?
- How do **lead time** and **lag time** affect a project schedule?
- What is the **critical path** and why is it important?

Case Study

- You are managing a **12-month project** to develop a new HR Management System for a mid-sized company.
 - The project scope includes requirements gathering, design, development, testing, training, and deployment.
 - The client insists that deployment **must** be completed by December to align with their new fiscal year.
 - Key constraints:
 - Only two senior developers are available at a time.
 - The design team is also working on another project and will be free only after 4 weeks.
 - Vendor approval for integration with payroll software may take up to 3 weeks.

Case study based Questions

- Identify **three possible scope risks** in this project and how you would address them.
- Suggest how a **Work Breakdown Structure (WBS)** could help control scope creep in this project.
- How would you ensure **scope validation** at each stage of the project?
- Which activities in this project are likely to be **mandatory dependencies**, and which might be **external dependencies**?
- If vendor approval takes **5 weeks instead of 3**, how might that impact the critical path?



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